Lecture 3 Compendium: Infrastructure Requirements & Challenges in Developing Contexts

AEC 813: Artificial Intelligence, IoT, and GIS Application in Agriculture

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Module 1: Introduction & Basics

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1 Introduction: The Infrastructure Triad

While Western agriculture focuses on "Optimization" (how to get 1% more yield), agriculture in the Global South (Developing Countries) focuses on "Resilience" and "Feasibility." Implementing IoT and GIS in Tanzania requires a fundamental re-architecting of the standard technology stack.

We analyze the system through the **Infrastructure Triad**:

1. **Connectivity:** How data moves (The Pipe).
2. **Power:** How devices stay alive (The Heart).
3. **Computation:** Where decisions are made (The Brain).

The Research Gap

Most academic literature assumes constant power and high-bandwidth LTE/5G coverage. Your PhD research must explicitly address the constraints of intermittent power, sparse connectivity, and severe environmental factors (heat/dust) found in tropical agro-ecologies.

2 Connectivity Architecture: Physics & Protocols

In rural environments, the primary constraint is the trade-off between **Range** and **Bandwidth**. This is governed by Information Theory.

2.1 The Shannon-Hartley Theorem

This theorem tells us the maximum rate at which information can be transmitted over a communications channel of a specified bandwidth in the presence of noise.

$$C = B \cdot \log_2 \left(1 + \frac{S}{N}\right) \tag{1}$$

Where:

- C is the Channel Capacity (bits per second).
- B is the Bandwidth of the channel (Hz).
- S/N is the Signal-to-Noise ratio.

Implication for Agriculture: To achieve long range (where Signal S is very weak), we must either:

1. Reduce the Capacity C (Send very little data).
2. Increase the Bandwidth B (Spread Spectrum techniques).

This is the fundamental principle behind **LoRa (Long Range)** technology. It uses Chirp Spread Spectrum (CSS) to trade data rate for receiver sensitivity.

2.2 LoRaWAN Architecture

For Tanzanian smallholders, Cellular (GSM) is expensive and power-hungry. LoRaWAN is the standard for rural IoT.

2.2.1 Spreading Factors (SF)

LoRa uses orthogonal spreading factors ($SF7$ to $SF12$).

- **SF7:** Short range, higher data rate, lower battery usage.
- **SF12:** Extreme range (>15km), very low data rate, high battery usage.

2.2.2 Mathematical Model: Time on Air (ToA)

To maximize battery life, we must minimize ToA. The duration of a packet depends on the Symbol Duration (T_{sym}).

$$T_{sym} = \frac{2^{SF}}{BW} \tag{2}$$

Where BW is Bandwidth (typically 125 kHz). Higher SF doubles the symbol duration, doubling the energy cost.

2.3 Python Simulation: LoRaWAN Link Optimization

We will simulate the "Time on Air" and energy consumption for different Spreading Factors to help you choose the right configuration for your deployment.

```python
import math

def calculate_lora_toa(sf, bandwidth_hz, payload_bytes, coding_rate=1):
    """
    Calculate Time on Air (ToA) for a LoRa packet.
    sf: Spreading Factor (7-12)
    bandwidth_hz: e.g., 125000
    payload_bytes: Application data size
    coding_rate: Error correction (1=4/5, 4=4/8)
    """
    # 1. Symbol Duration
    t_symbol = (2**sf) / bandwidth_hz

    # 2. Preamble Duration (Standard 8 symbols)
    n_preamble = 8
    t_preamble = (n_preamble + 4.25) * t_symbol

    # 3. Payload Symbol Count (LoRa Packet Formula)
    # Using explicit LoRa formula denominator logic
    term1 = 8 * payload_bytes - 4 * sf + 28 + 16 # Header overhead
    term2 = 4 * (sf - 2 * 0) # Implicit header mode disabled
    n_payload = 8 + max(math.ceil(term1 / term2) * (coding_rate + 4), 0)

    t_payload = n_payload * t_symbol

    return t_preamble + t_payload

def energy_consumption_simulation():
    # Hardware constants (SX1276 Chip)
    voltage = 3.3 # Volts
    current_tx = 120e-3 # Amps (120mA) during TX

    payload = 50 # Bytes (e.g., Soil Moisture + Temp + Battery Level)
    bandwidth = 125000 # 125 kHz

    print(f"{'SF':<5} | {'ToA (ms)':<10} | {'Energy (mJ)':<15} | {'Max Packets/Day'}")
    print("-" * 55)
```

```
38
39      for sf in range(7, 13):
40          toa_sec = calculate_lora_toa(sf, bandwidth, payload)
41          toa_ms = toa_sec * 1000
42
43          # Energy = Voltage * Current * Time
44          energy_joules = voltage * current_tx * toa_sec
45          energy_mj = energy_joules * 1000
46
47          # Duty Cycle Limit (1% in ISM Band)
48          # Max TX time per day = 864 seconds (1% of 24h)
49          max_packets = int(864 / toa_sec)
50
51          print(f"{sf:<5} | {toa_ms:<10.1f} | {energy_mj:<15.2f} | {max_packets}")
52
53  energy_consumption_simulation()
```

Listing 1: Calculating LoRaWAN Time-on-Air and Energy Cost

Analysis: At SF12, a single packet might take 1-2 seconds to transmit. This consumes significant energy. Your research architecture must balance range (SF) against battery life.

3 Energy Architecture: Thermodynamics & Harvesting

In the Global South, "The Grid" is unreliable or non-existent on farmlands. Batteries are finite. We must design for **Energy Neutrality**.

$$E_{harvested} \geq E_{consumed} + E_{leakage} \tag{3}$$

3.1 Battery Chemistry in Tropical Climates

Common Lithium-Ion (Li-Ion) batteries degrade rapidly above 40°C. In Morogoro/Dodoma, soil surface temperatures can exceed 50°C.

- **Li-Ion / Li-Po:** High density, but dangerous in heat. Max temp ≈ 60°C.
- **LiFePO4 (LFP):** Lower density, but very stable. Excellent for solar cycling. Best choice for AgTech.
- **LTO (Lithium Titanate):** Extreme lifespan (20k cycles) and temp range, but expensive.

3.2 Solar Harvesting Dimensioning

To dimension a solar panel for an IoT node, we calculate the **Daily Energy Budget**.

$$E_{daily} = (P_{active} \cdot t_{active}) + (P_{sleep} \cdot t_{sleep})$$

Where:

- P_{active}: Power during sensing/transmission ($\approx$ 400mW).
- t_{active}: Total time active (e.g., 2 seconds * 24 times = 48s).
- P_{sleep}: Power during deep sleep ($\approx$ 0.05mW).
- t_{sleep}: Remainder of the day.

Autonomy Factor: In Tanzania, we design for the "Long Rains" (Masika), assuming 3-5 days of heavy cloud cover (No solar).

$$C_{batt} \geq \frac{E_{daily} \cdot Days_{autonomy}}{V_{batt} \cdot DoD}$$

Where DoD is Depth of Discharge (0.8 for Li-Ion).

3.3 Python Simulation: Solar Year Simulation

We simulate a battery's charge level over a year using NASA POWER weather data (simulated here) to see if the node dies during the rainy season.

```python
import numpy as np
import matplotlib.pyplot as plt

def solar_viability_simulation():
    # --- 1. Load/Simulate Weather Data ---
    days = 365
    # Simulate Peak Sun Hours (PSH) for Tanzania
    # High in Dry Season (Jan-Feb, Jun-Oct), Low in Rainy (Mar-May, Nov-Dec)
    t = np.linspace(0, 4*np.pi, days)
    # Base 5 hours +/- 2 hours fluctuation + random cloud noise
    sun_hours = 5 + 2 * np.sin(t) + np.random.normal(0, 1, days)
    sun_hours = np.clip(sun_hours, 1, 8) # Clip between 1 and 8 hours

    # --- 2. System Specs ---
    panel_wattage = 1.0 # 1 Watt small panel
    panel_efficiency = 0.7 # Dust, heat losses
    battery_capacity_wh = 10.0 # 10 Watt-hours (~2500mAh @ 3.7V)

    daily_consumption_wh = 1.5 # High consumption node

    # --- 3. Simulation Loop ---
    battery_state = 5.0 # Start half full
    battery_history = []

    for day in range(days):
        # Energy In
        energy_in = panel_wattage * sun_hours[day] * panel_efficiency

        # Energy Balance
        battery_state += energy_in
        battery_state -= daily_consumption_wh

        # Clamping (Cannot exceed capacity, cannot drop below 0)
        if battery_state > battery_capacity_wh:
            battery_state = battery_capacity_wh
        if battery_state < 0:
            battery_state = 0 # SYSTEM FAILURE

        battery_history.append(battery_state)

    # --- 4. Plotting ---
    plt.figure(figsize=(12, 6))
    plt.plot(battery_history, label='Battery Charge (Wh)')
    plt.axhline(y=0, color='red', linestyle='--', label='System Failure')
    plt.title('1-Year Energy Simulation (Tanzania Climate Profile)')
    plt.xlabel('Day of Year')
    plt.ylabel('Battery Energy (Wh)')
    plt.legend()
    plt.grid(True)
    plt.show()
```

```
51
52      failures = battery_history.count(0)
53      print(f"System Failed (Dead Battery) on {failures} days.")
54
55  solar_viability_simulation()
```

Listing 2: Solar Energy Balance Simulation

4 Computing Infrastructure: Edge vs. Cloud

In developed nations, data is cheap. In Tanzania, data bundles are a significant operational expenditure (OPEX) for farmers.

4.1 The Cost of Data Transmission

Research (e.g., by Vuran et al.) shows that transmitting 1 bit of data wirelessly consumes roughly the same energy as executing 1,000 - 5,000 instructions on a local microcontroller.

Architectural Rule: Compute locally, transmit sparsely.

- **Raw Data Architecture:** Send image of leaf to cloud → Cloud AI analyzes → Result back. (High Bandwidth, High Energy).
- **Edge Architecture:** Analyze leaf on device → Send result "0" (Healthy) or "1" (Sick). (Low Bandwidth, Low Energy).

4.2 Dataset: NASA POWER

For your infrastructure planning, do not guess weather data. Use the **NASA POWER (Prediction Of Worldwide Energy Resources)** open API. It provides solar irradiance and meteorology data for any lat/long.

- URL: `https://power.larc.nasa.gov/`
- Use Case: Calculating exact solar panel size for a node in Iringa vs. Dar es Salaam.

5 Socio-Technical Challenges

A PhD in Agroecology is not just engineering; it is social science.

5.1 Theft and Vandalism

High-value components (Solar panels, Batteries) are targets in unfenced fields.

- **Mitigation:** "Camouflaged IoT." designing sensors that look like rocks or sticks.
- **Community Engagement:** Employing local "Data Guardians."

5.2 The Maintenance Gap

The "Pilotitis" disease: Projects succeed during the PhD (while you are there to fix wires) but fail 6 months after you graduate.

- **Design Constraint:** Systems must be modular and repairable using locally available parts (e.g., replacing a specialized LiPo with standard 18650 cells found in laptop batteries).

5.3 Techno-Economic Analysis (CAPEX vs OPEX)

Farmers care about ROI.

$$ROI = \frac{\text{Value of Yield Increase} - (\text{CAPEX} + \text{OPEX})}{\text{CAPEX} + \text{OPEX}} \tag{4}$$

If a LoRaWAN gateway costs \$200 but saves \$20 of water per year, the ROI period is 10 years. This is unacceptable for smallholders. We must aim for ROI < 2 years.

6 Review Questions & Assignment

1. **Design Challenge:** Design an IoT node for a remote Rice Paddy in Kilombero.
   - Constraint: No cellular signal.
   - Constraint: Rainy season lasts 3 months.
   - Requirement: Measure water level every 30 minutes.
   - Task: Choose the Protocol, Battery Chemistry, and Solar Panel size. Justify with calculations.
2. **Math:** Using the Shannon-Hartley theorem, explain why LoRaWAN has such a low data rate compared to WiFi, given they occupy similar bandwidths (125kHz vs 20MHz).
3. **Coding:** Modify the Python Solar Simulation to include a "Deep Sleep" logic. If the battery drops below 30%, stop sampling until it recharges to 50%. How many "Data Gaps" does this create?

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